

Math 12 - Intermediate Algebra

Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e

COURSE AT A GLANCE

Review Content

REVIEW

1.1 Use the Language of Algebra; 1.2 Integers; 1.3 Fractions; 1.4 Decimals; 1.5 Properties of Real Numbers

UNIT LEARNING OBJECTIVES

- I can evaluate numerical expressions using the order of operations.
- I can perform operations with signed integers and fractions.
- I can simplify an algebraic expression by combining like terms.
- I can use the distributive property to rewrite an expression.
- I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

UNIT LEARNING OBJECTIVES

- I can solve a linear equation and present equivalent steps clearly.
- I can solve a linear equation containing fractions.
- I can rearrange a formula to isolate a specified variable.
- I can translate a verbal relationship into a linear equation.
- I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

UNIT LEARNING OBJECTIVES

- I can solve and graph a one-variable linear inequality.
- I can solve a compound inequality involving AND.
- I can solve a compound inequality involving OR.
- I can express an inequality solution using interval notation.
- I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation in a common form.
- I can write an equation of a line from given information.
- I can determine whether two lines are parallel, perpendicular, or neither.
- I can evaluate a function using function notation.
- I can determine whether a relation is a function using the vertical line test or input-output data.
- I can determine domain and range from a graph or table.
- I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

UNIT LEARNING OBJECTIVES

- I can interpret the solution of a linear system as an intersection point.
- I can solve a two-variable linear system by graphing.
- I can solve a two-variable linear system by substitution.
- I can solve a two-variable linear system by elimination.
- I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

UNIT LEARNING OBJECTIVES

- I can simplify expressions using integer exponent rules.
- I can add and subtract polynomials.
- I can multiply polynomials, including binomials.
- I can divide a polynomial by a monomial.
- I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

2.7 Solve Absolute Value Inequalities; 3.4 Graph Linear Inequalities in Two Variables; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

- I can solve and graph an absolute-value inequality.
- I can graph a system of linear inequalities and identify its solution region.
- I can factor a quadratic trinomial.
- I can identify a relationship as discrete or continuous.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

Textbook coverage: 1.1 Use the Language of Algebra; 1.2 Integers; 1.3 Fractions; 1.4 Decimals; 1.5 Properties of Real Numbers

M12-REV-001

I can evaluate numerical expressions using the order of operations.

Textbook: 1.1 Use the Language of Algebra

SUPPORTING SKILLS



Apply the order of operations to a numerical expression.

SK-NUM-ORDER-OPERATIONS · inherited from middle school

M12-REV-002

I can perform operations with signed integers and fractions.

Textbook: 1.2 Integers; 1.3 Fractions; 1.4 Decimals

SUPPORTING SKILLS



Add, subtract, multiply, and divide fractions.

SK-NUM-FRACTION-OPERATE · inherited from middle school



Add, subtract, multiply, and divide signed integers.

SK-NUM-INTEGER-OPERATE · inherited from middle school

M12-REV-003

I can simplify an algebraic expression by combining like terms.

Textbook: 1.5 Properties of Real Numbers

SUPPORTING SKILLS

- R** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- R** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school

M12-REV-004

I can use the distributive property to rewrite an expression.

Textbook: 1.5 Properties of Real Numbers

SUPPORTING SKILLS

- R** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school

M12-REV-005

I can determine the prime factorization of a whole number.

Textbook: 1.1 Use the Language of Algebra

SUPPORTING SKILLS

- R** Express a whole number as a product of prime factors.
SK-NUM-PRIME-FACTOR · inherited from middle school

Linear Equations and Applications

REQUIRED

Textbook coverage: 2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

M12-EQN-001

I can solve a linear equation and present equivalent steps clearly.

Textbook: 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

- I** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION
 - A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
 - I** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS
 - A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
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M12-EQN-002

I can solve a linear equation containing fractions.

Textbook: 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
 - I** Clear fractions from an equation using a common denominator.
SK-ALG-CLEAR-FRACTIONS
 - A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
 - A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
-

M12-EQN-003

I can rearrange a formula to isolate a specified variable.

Textbook: 2.3 Solve a Formula for a Specific Variable

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
 - I** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE
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M12-EQN-004

I can translate a verbal relationship into a linear equation.

Textbook: 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION

I can solve and interpret a number, mixture, or motion problem using a linear equation.

Textbook: 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

Inequalities and Absolute-Value Equations

REQUIRED

Textbook coverage: 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

M12-INE-001

I can solve and graph a one-variable linear inequality.

Textbook: 2.5 Solve Linear Inequalities

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE
- I** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN

M12-INE-002

I can solve a compound inequality involving AND.

Textbook: 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Interpret AND as the intersection of two inequality solution sets.
SK-INE-COMPOUND-AND
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- A** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-003

I can solve a compound inequality involving OR.

Textbook: 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Interpret OR as the union of two inequality solution sets.
SK-INE-COMPOUND-OR
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- A** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-004

I can express an inequality solution using interval notation.

Textbook: 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- I** Translate between inequality and interval notation.
SK-NOT-INTERVAL

I can solve an absolute-value equation and check its solutions.

Textbook: 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

- I** Rewrite an absolute-value equation as two cases.

SK-ABS-SPLIT-EQUATION

- A** Check a proposed solution in the original equation.

SK-ALG-CHECK-SOLUTION · first introduced in Math 12

- A** Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

Lines and Introductory Functions

REQUIRED

Textbook coverage: 3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions; 3.6 Graphs of Functions

M12-LIN-001

I can determine and interpret the slope of a line.

Textbook: 3.2 Slope of a Line

SUPPORTING SKILLS

- I** Determine slope from a graph.
SK-LIN-SLOPE-GRAPH
- I** Calculate slope from two points.
SK-LIN-SLOPE-POINTS
- I** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS

M12-LIN-002

I can graph a line from an equation in a common form.

Textbook: 3.1 Graph Linear Equations in Two Variables

SUPPORTING SKILLS

- I** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT
- I** Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE
- I** Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT

M12-LIN-003

I can write an equation of a line from given information.

Textbook: 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- A** Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M12-LIN-004

I can determine whether two lines are parallel, perpendicular, or neither.

Textbook: 3.2 Slope of a Line; 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- I** Recognize that distinct parallel lines have equal slopes.
SK-LIN-PARALLEL-SLOPE
- I** Determine the negative reciprocal of a nonzero slope.
SK-LIN-PERPENDICULAR-SLOPE

M12-LIN-005

I can evaluate a function using function notation.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT
- I** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ

M12-LIN-006

I can determine whether a relation is a function using the vertical line test or input-output data.

Textbook: 3.5 Relations and Functions

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE

M12-LIN-008

I can determine domain and range from a graph or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- I** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH
- I** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE

M12-LIN-009

I can interpret domain and range in context.

Textbook: 3.6 Graphs of Functions

SUPPORTING SKILLS

- I** Restrict domain to values meaningful in a context.
SK-FUNC-CONTEXT-DOMAIN
- I** Interpret the range of a function in a context.
SK-FUNC-CONTEXT-RANGE
- I** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN

Systems of Linear Equations

REQUIRED

Textbook coverage: 4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

M12-SYS-001

I can interpret the solution of a linear system as an intersection point.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- I Locate and interpret the intersection of two lines.

SK-SYS-GRAPH-INTERSECTION

M12-SYS-002

I can solve a two-variable linear system by graphing.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Use slope to generate additional points on a line.

SK-LIN-GRAPH-SLOPE · first introduced in Math 12

- A Plot an intercept accurately on a coordinate plane.

SK-LIN-PLOT-INTERCEPT · first introduced in Math 12

- I Check an ordered pair in both equations of a system.

SK-SYS-CHECK

- D Locate and interpret the intersection of two lines.

SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M12-SYS-003

I can solve a two-variable linear system by substitution.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

- A Check an ordered pair in both equations of a system.

SK-SYS-CHECK · first introduced in Math 12

- I Substitute an equivalent expression for a variable.

SK-SYS-SUBSTITUTE

M12-SYS-004

I can solve a two-variable linear system by elimination.

Textbook: 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

- A Check an ordered pair in both equations of a system.

SK-SYS-CHECK · first introduced in Math 12

- I Add equations to eliminate a variable.

SK-SYS-ELIMINATE

I can create and solve a two-variable linear system for an application.

Textbook: 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

SUPPORTING SKILLS

- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- I** Define two variables and write two equations for a context.
SK-SYS-MODEL
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

Polynomial Operations and Basic Factoring

REQUIRED

Textbook coverage: 5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

M12-POL-001

I can simplify expressions using integer exponent rules.

Textbook: 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- I** Apply the power rule for exponents.
SK-EXP-POWER
- I** Apply the product rule for exponents.
SK-EXP-PRODUCT
- I** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT

M12-POL-002

I can add and subtract polynomials.

Textbook: 5.1 Add and Subtract Polynomials

SUPPORTING SKILLS

- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
- I** Identify like polynomial terms by variable part and exponent.
SK-POLY-LIKE-TERMS
- I** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY

M12-POL-003

I can multiply polynomials, including binomials.

Textbook: 5.3 Multiply Polynomials

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY
- I** Multiply a monomial by a polynomial.
SK-POLY-DISTRIBUTE
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-004

I can divide a polynomial by a monomial.

Textbook: 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- I** Divide each term of a polynomial by a monomial.
SK-POLY-DIVIDE-MONOMIAL
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-005

I can factor a greatest common factor from a polynomial.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Factor the greatest common monomial factor from a polynomial.
SK-FACTOR-GCF
- I** Determine the numerical and variable greatest common factor of polynomial terms.
SK-POLY-GCF
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

Extension Content

EXTENSION

Textbook coverage: 2.7 Solve Absolute Value Inequalities; 3.4 Graph Linear Inequalities in Two Variables; 3.6 Graphs of Functions

M12-EXT-001

I can solve and graph an absolute-value inequality.

Textbook: 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

- I** Rewrite an absolute-value inequality as a compound inequality.
SK-ABS-SPLIT-INEQUALITY
- A** Interpret AND as the intersection of two inequality solution sets.
SK-INE-COMPOUND-AND · first introduced in Math 12
- A** Interpret OR as the union of two inequality solution sets.
SK-INE-COMPOUND-OR · first introduced in Math 12
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12

M12-EXT-002

I can graph a system of linear inequalities and identify its solution region.

Textbook: 3.4 Graph Linear Inequalities in Two Variables

SUPPORTING SKILLS

- I** Graph the boundary line of a linear inequality with the correct line type.
SK-INE-GRAPH-BOUNDARY
- I** Identify the overlapping solution region of multiple linear inequalities.
SK-INE-SYSTEM-REGION
- A** Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE · first introduced in Math 12

M12-EXT-003

I can factor a quadratic trinomial.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I** Verify a factorization by multiplication.
SK-FACTOR-CHECK
- I** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL
- I** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC

M12-EXT-004

I can identify a relationship as discrete or continuous.

Textbook: 3.6 Graphs of Functions

SUPPORTING SKILLS

- I** Distinguish discrete inputs from values varying continuously over an interval.
SK-FUNC-DISCRETE-CONTINUOUS